OPERATIONS MANAGEMENT

CONSTRAINT MANAGEMENT

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Constraint Management

- A **constraint** is any factor that limits the performance of a system and restricts its output (capacity).

- When constraints exist at any step, capacity can become imbalanced and the overall performance of the system suffers.

- Constraints can occur up or down the supply chain, with either the firm's suppliers or customers, or within one of the firm's processes like service/product development, or order fulfillment.

- Three kinds of constraints can generally be identified: **physical** (usually machine, labor, or workstation capacity or material shortages, but could be space or quality), **market** (demand is less than capacity), or **managerial** (policy, metrics, or mind-sets that create constraints that impede work flow).

- A **bottleneck** is a special type of a constraint that relates to the capacity shortage of a process, and is defined as any resource whose available capacity limits the organization's ability to meet the service or product volume, product mix, or fluctuating requirements demanded by the marketplace.

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- Under certain conditions, a bottleneck is also called a **capacity constrained resource (CCR)**. The process with the least capacity is called a bottleneck if its output is less than the market demand, or called a CCR if it is the least capable resource in the system but still has higher capacity than the market demand.

- Firms must manage their constraints and make appropriate capacity choices at the individual-process level, as well as at the organization level. Detailed decisions and choices made within each of these levels affect where resource constraints or bottlenecks show up. Relieving a bottleneck in one part of an organization might not have the desired effect unless a bottleneck in another part of the organization is also addressed.

- A bottleneck could be the sales department not getting enough sales or the loan department not processing loans fast enough. The constraint could be a lack of capital or equipment, or it could be planning and scheduling.

- Managers throughout the organization must understand how to identify and manage bottlenecks in all types of processes, how to relate the capacity and performance measures of one process to another, and how to use that information to determine the firm's best service or product mix.

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The Theory of Constraints

- The **theory of constraints (TOC)** is a systematic management approach that focuses on actively managing those constraints that impede a firm's progress toward its goal of maximizing profits and effectively using its resources. It outlines a deliberate process for identifying and overcoming constraints. The process focuses not just on the efficiency of individual processes, but also on the bottlenecks that constrain the system as a whole.

- TOC methods increase the firm's profits more effectively by focusing on making materials flow rapidly through the entire system. They help firms look at the big picture — how processes can be improved to increase overall work flows, and how inventory and workforce levels can be reduced while still effectively utilizing critical resources.

- To do this, it is important to understand the relevant performance and capacity measures at the operational level, as well as their relationship to the more broadly understood financial measures at the firm level.

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TABLE 7.1 HOW THE FIRM'S OPERATIONAL MEASURES RELATE TO ITS FINANCIAL MEASURES

Operational Measures	TOC View	Relationship to Financial Measures
Inventory (I)	All the money invested in a system in purchasing things that it intends to sell	A decrease in I leads to an increase in net profit, ROI, and cash flow.
Throughput (T)	Rate at which a system generates money through sales	An increase in T leads to an increase in net profit, ROI, and cash flows.
Operating Expense (OE)	All the money a system spends to turn inventory into throughput	A decrease in OE leads to an increase in net profit, ROI, and cash flows.
Utilization (U)	The degree to which equipment, space, or workforce is currently being used, and is measured as the ratio of average output rate to maximum capacity, expressed as a percentage	An increase in U at the bottleneck leads to an increase in net profit, ROI, and cash flows.

- Producing a product or a service that does not lead to a sale will not increase a firm's throughput, but will increase its inventory and operating expenses.

- It is always best to manage the system so that utilization at the bottleneck resource is maximized in order to maximize throughput.

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Key Principles of The TOC

TABLE 7.2 | SEVEN KEY PRINCIPLES OF THE THEORY OF CONSTRAINTS

1. **The focus should be on balancing flow, not on balancing capacity.**
2. **Maximizing the output and efficiency of every resource may not maximize the throughput of the entire system.**
3. **An hour lost at a bottleneck or a constrained resource is an hour lost for the whole system. In contrast, an hour saved at a nonbottleneck resource is a mirage because it does not make the whole system more productive.**
4. **Inventory is needed only in front of the bottlenecks in order to prevent them from sitting idle, and in front of assembly and shipping points in order to protect customer schedules. Building inventories elsewhere should be avoided.**
5. **Work, which can be materials, information to be processed, documents, or customers, should be released into the system only as frequently as the bottlenecks need it. Bottleneck flows should be equal to the market demand. Pacing everything to the slowest resource minimizes inventory and operating expenses.**
6. **Activating a nonbottleneck resource (using it for improved efficiency that does not increase throughput) is not the same as utilizing a bottleneck resource (that does lead to increased throughput). Activation of nonbottleneck resources cannot increase throughput, nor promote better performance on financial measures outlined in Table 7.1.**
7. **Every capital investment must be viewed from the perspective of its global impact on overall throughput (T), inventory (I), and operating expense (OE).**

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Practical application of the TOC involves the implementation of the following steps:

1. Identify the System Bottleneck(s)

2. Exploit the Bottleneck(s)

3. Subordinate All Other Decisions to Step 2

4. Elevate the Bottleneck(s)

5. Do Not Let Inertia Set In

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Identification and Management of Bottlenecks

- Bottlenecks can both be internal or external to the firm, and typically represent a process, a step, or a workstation with the lowest capacity.

- **Throughput time** is the total elapsed time from the start to the finish of a job or a customer being processed at one or more work centers.

- Where a bottleneck lies in a given service or manufacturing process can be identified in two ways.

- A workstation in a process is a bottleneck if **(1)** it has the highest total time per unit processed, or **(2)** it has the highest average utilization and total workload.

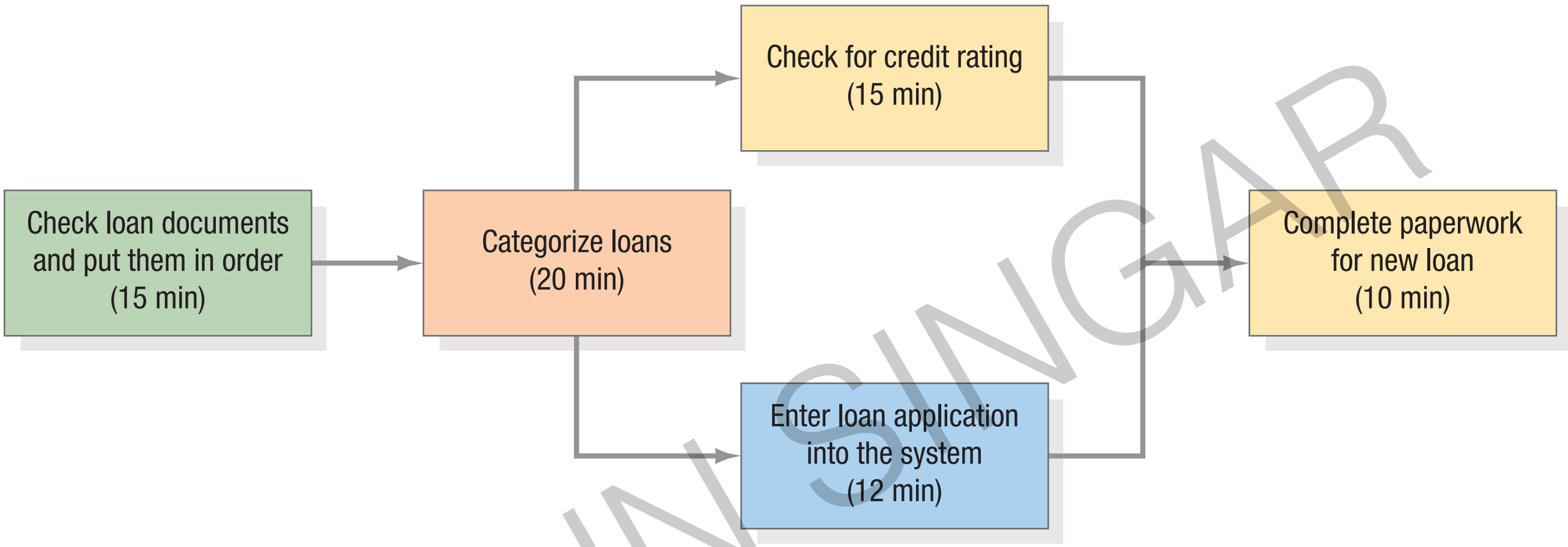
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Managing Bottlenecks in Service Processes

Example

- Managers at the First Community Bank are attempting to shorten the time it takes customers with approved loan applications to get their paperwork processed. The flowchart for this process, consisting of several different activities, each performed by a different bank employee, is shown in the Figure. Approved loan applications first arrive at activity or step 1, where they are checked for completeness and put in order. At step 2, the loans are categorized into different classes according to the loan amount and whether they are being requested for personal or commercial reasons. While credit checking commences at step 3, loan application data are entered in parallel into the information system for record-keeping purposes at step 4. Finally, all paperwork for setting up the new loan is finished at step 5. The time taken in minutes is given in parentheses.

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Which single step is the bottleneck, assuming that market demand for loan applications exceeds the capacity of the process? The management is also interested in knowing the maximum number of approved loans this system can process in a 5-hour work day.

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Solution

- We define the bottleneck as step 2, which has the highest time per loan processed. **The throughput time to complete an approved loan application is 15 + 20 + max (15, 12) + 10 = 60 minutes**. Although we assume no waiting time in front of any step, in practice such a smooth process flow is not always the case. So the actual time taken for completing an approved loan will be longer than 60 minutes due to nonuniform arrival of applications, variations in actual processing times, and the related factors.

- The capacity for loan completions is derived by translating the "minutes per customer" at the bottleneck step to "customer per hour." At First Community Bank, **it is 3 customers per hour because the bottleneck step 2 can process only 1 customer every 20 minutes**.

Decision Point

- Step 2 is the bottleneck constraint. **The bank will be able to complete a maximum of only 3 loan accounts per hour, or 15 new loan accounts in a 5-hour day**. Management can increase the flow of loan applications by increasing the capacity of step 2 up to the point where another step becomes the bottleneck.

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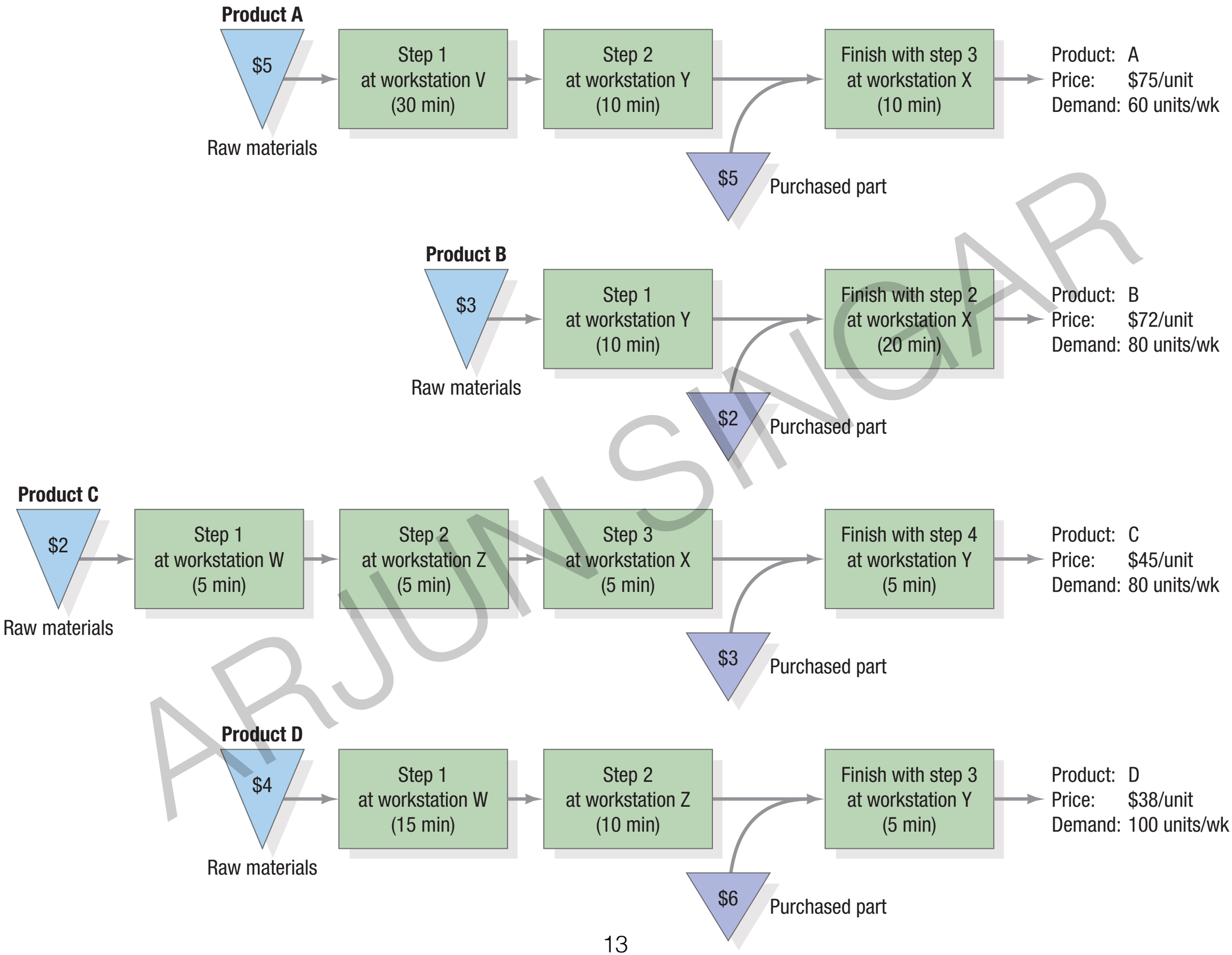
Managing Bottlenecks in Manufacturing Processes

Example

- Diablo Electronics manufactures four unique products (A, B, C, and D) that are fabricated and assembled in five different workstations (V, W, X, Y, and Z) using a small batch process. Each workstation is staffed by a worker who is dedicated to work a single shift per day at an assigned workstation. Batch setup times have been reduced to such an extent that they can be considered negligible. A flowchart denotes the path each product follows through the manufacturing process as shown in the Figure, where each product's price, demand per week, and processing times per unit are indicated as well. Inverted triangles represent purchased parts and raw materials consumed per unit at different workstations. Diablo can make and sell up to the limit of its demand per week, and no penalties are incurred for not being able to meet all the demand.

- Which of the five workstations (V, W, X, Y, or Z) has the highest utilization, and thus serves as the bottleneck for Diablo Electronics?

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Solution

- Because the denominator in the utilization ratio is the same for every workstation, with one worker per machine at each step in the process, we can simply identify the bottleneck by computing aggregate workloads at each workstation.

- The firm wants to satisfy as much of the product demand in a week as it can. Each week consists of 2,400 minutes of available production time. Multiplying the processing time at each station for a given product with the number of units demanded per week yields the workload represented by that product. These loads are summed across all products going through a workstation to arrive at the total load for the workstation, which is then compared with the others and the existing capacity of 2,400 minutes.

Workstation	Load from Product A	Load from Product B	Load from Product C	Load from Product D	Total Load (min)
V	60 × 30 = 1800	0	0	0	1,800
W	0	0	80 × 5 = 400	100 × 15 = 1,500	1,900
X	60 × 10 = 600	80 × 20 = 1,600	80 × 5 = 400	0	2,600
Y	60 × 10 = 600	80 × 10 = 800	80 × 5 = 400	100 × 5 = 500	2,300
Z	0	0	80 × 5 = 400	100 × 10 = 1,000	1,400

Decision Point

Workstation X is the bottleneck for Diablo Electronics because the aggregate workload at X is larger than the aggregate workloads of workstations V, W, Y, and Z and the maximum available capacity of 2,400 minutes per week.

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Relieving Bottlenecks

- Managers should minimize idle time at the bottlenecks, caused by delays elsewhere in the system and make sure that the bottleneck has all the resources it needs to stay busy.

- When a changeover or setup is made at a bottleneck, the number of units or customers processed before the next changeover should be large compared to the number processed at less critical operations.

- Maximizing the number of units processed per setup means fewer setups per year and, thus, less total time lost to setups.

- The long-term capacity of bottleneck operations can be expanded in various ways. Investments can be made in new equipment and in brick-and-mortar facility expansions. The bottleneck's capacity also can be expanded by operating it more hours per week, such as by hiring more employees and going from a one-shift operation to multiple shifts, or by hiring more employees and operating the plant 6 or 7 days per week versus 5 days per week. Managers also might relieve the bottleneck by redesigning the process, either through *process reengineering* or *process improvement*, or by purchasing additional machines or machines that can handle more capacity.

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Applying the TOC to Product Mix Decisions

- Managers might be tempted to produce the products with the highest contribution margins or unit sales.

- *Contribution margin* is the amount each product contributes to profits and overhead; no fixed costs are considered when making the product mix decision. We call this approach the *traditional method.*

- The problem with this approach is that the firm's actual throughput and overall profitability depend more upon the contribution margin generated at the bottleneck than by the contribution margin of each individual product produced. We call this latter approach the *bottleneck method*.

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Determining the Product Mix Using Contribution Margin

Example

- The senior management at Diablo Electronics (see previous example) wants to improve profitability by accepting the right set of orders, and so collected some additional financial data. Variable overhead costs are $8,500 per week. Each worker is paid $18 per hour and is paid for an entire week, regardless of how much the worker is used. Consequently, labor costs are fixed expenses. The plant operates one 8-hour shift per day, or 40 hours each week. Currently, decisions are made using the traditional method, which is to accept as much of the highest contribution margin product as possible (up to the limit of its demand), followed by the next highest contribution margin product, and so on until no more capacity is available. Pedro Rodriguez, the newly hired production supervisor, is knowledgeable about the theory of constraints and bottleneck-based scheduling. He believes that profitability can indeed be improved if bottleneck resources were exploited to determine the product mix. What is the change in profits if, instead of the traditional method used by Diablo Electronics, the bottleneck method advocated by Pedro is used to select the product mix?

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SOLUTION

Decision Rule 1: Traditional Method

Select the best product mix according to the highest overall contribution margin of each product.

Step 1: Calculate the contribution margin per unit of each product as shown here.

	A	B	C	D
Price	$75.00	$72.00	$45.00	$38.00
Raw material and purchased parts	–10.00	–5.00	–5.00	–10.00
= Contribution margin	$65.00	$67.00	$40.00	$28.00

When ordered from highest to lowest, the contribution margin per unit sequence of these products is B, A, C, D.

Step 2: Allocate resources V, W, X, Y, and Z to the products in the order decided in step 1. Satisfy each demand until the bottleneck resource (workstation X) is encountered. Subtract minutes away from 2,400 minutes available for each week at each stage.

Work Center	Minutes at the Start	Minutes Left After Making 80 B	Minutes Left After Making 60 A	Can Only Make 40 C	Can Still Make 100 D
V	2,400	2,400	600	600	600
W	2,400	2,400	2,400	2,200	700
X	2,400	800	200	0	0
Y	2,400	1,600	1,000	800	300
Z	2,400	2,400	2,400	2,200	1,200

The best product mix according to this traditional approach is then 60 A, 80 B, 40 C, and 100 D.

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Step 3: Compute profitability for the selected product mix.

Profits		
Revenue	$(60 \times \$75) + (80 \times \$72) + (40 \times \$45) + (100 \times \$38)$	= \$15,860
Materials	$(60 \times \$10) + (80 \times \$5) + (40 \times \$5) + (100 \times \$10)$	= –\$2,200
Labor	(5 workers) × (8 hours/day) × (5 days/week) × (\$18/hour)	= –\$3,600
Overhead		= –\$8,500
Profit		= \$1,560

Manufacturing the product mix of 60 A, 80 B, 40 C, and 100 D will yield a profit of \$1,560 per week.

Decision Rule 2: Bottleneck Method

Select the best product mix according to the dollar contribution margin per minute of processing time at the bottleneck workstation X. This method would take advantage of the principles outlined in the theory of constraints and get the most dollar benefit from the bottleneck.

Step 1: Calculate the contribution margin/minute of processing time at bottleneck workstation X:

	Product A	Product B	Product C	Product D
Contribution margin	\$65.00	\$67.00	\$40.00	\$28.00
Time at bottleneck	10 minutes	20 minutes	5 minutes	0 minutes
Contribution margin per minute	\$6.50	\$3.35	\$8.00	Not defined

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When ordered from highest to lowest contribution margin/minute at the bottleneck, the manufacturing sequence of these products is D, C, A, B, which is reverse of the earlier order. Product D is scheduled first because it does not consume any resources at the bottleneck.

Step 2: Allocate resources V, W, X, Y, and Z to the products in the order decided in step 1. Satisfy each demand until the bottleneck resource (workstation X) is encountered. Subtract minutes away from 2,400 minutes available for each week at each stage.

Work Center	Minutes at the Start	Minutes Left After Making 100 D	Minutes Left After Making 80 C	Minutes Left After Making 60 A	Can Only Make 70 B
V	2,400	2,400	2,400	600	600
W	2,400	900	500	500	500
X	2,400	2,400	2,000	1,400	0
Y	2,400	1,900	1,500	900	200
Z	2,400	1,400	1,000	1,000	1,000

The best product mix according to this bottleneck-based approach is then 60 A, 70 B, 80 C, and 100 D.

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Step 3: Compute profitability for the selected product mix.

Profits		
Revenue	$(60 \times \$75) + (70 \times \$72) + (80 \times \$45) + (100 \times \$38)$	$= \$16{,}940$
Materials	$(60 \times \$10) + (70 \times \$5) + (80 \times \$5) + (100 \times \$10)$	$= -\$2{,}350$
Labor	(5 workers) × (8 hours/day) × (5 days/week) × ($18/hour)	$= -\$3{,}600$
Overhead		$= -\$8{,}500$
Profit		$= \$2{,}490$

Manufacturing the product mix of 60 A, 70 B, 80 C, and 100 D will yield a profit of $2,490 per week.

DECISION POINT

By focusing on the bottleneck resources in accepting customer orders and determining the product mix, the sequence in which products are selected for production is reversed from **B, A, C, D** to **D, C, A, B**. Consequently, the product mix is changed from 60 A, 80 B, 40 C, and 100 D to 60 A, 70 B, 80 C, and 100 D. The increase in profits by using the bottleneck method is $930, ($2,490 − $1,560), or almost 60 percent over the traditional approach.

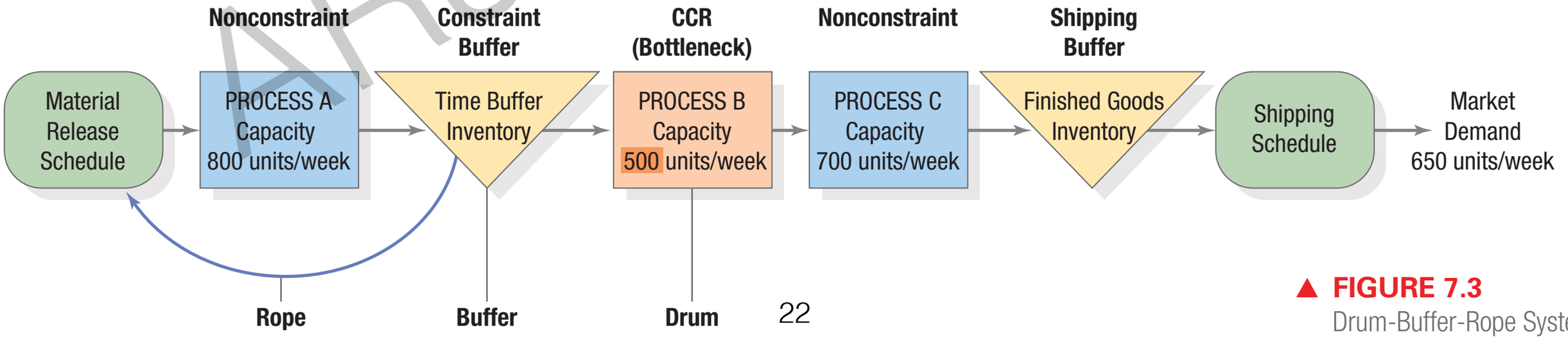
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Drum Buffer Rope Systems

- **Drum-Buffer-Rope (DBR)** is a planning and control system based on the theory of constraints that is often used in manufacturing firms to plan and schedule production. It works by regulating the flow of work-in-process materials at the bottleneck or the capacity constrained resource (CCR).

- The bottleneck schedule is the *drum* because it sets the beat or the production rate for the entire plant and is linked to the market demand. The *buffer* is a time buffer that plans early flows to the bottleneck and thus protects it from disruption. It also ensures that the bottleneck is never starved for work. Finally, the *rope* represents the tying of material release to the drum beat, which is the rate at which the bottleneck controls the throughput of the entire plant.

- To better understand the drum-buffer-rope system, consider the schematic layout shown in Figure 7.3. Process B, with a capacity of only 500 units per week, is the bottleneck because the up-stream Process A and downstream Process C have capacities of 800 units per week and 700 units per week, respectively, and the market demand is 650 units per week, on average. In this case, because the capacity at process B is less than the market demand, it is the bottleneck. A constraint time buffer, which can be in the form of materials arriving earlier than needed, is placed right in front of the bottleneck (Process B). A shipping buffer, in the form of finished goods inventory, can also be placed prior to the shipping schedule in order to protect customer orders that are firm. Finally, a rope ties the material release schedule to match the schedule, or drum beat, at the bottleneck. The material flow is pulled forward by the drum beat prior to the bottleneck, while it is pushed downstream toward the customer subsequent to the bottleneck.



▲ **FIGURE 7.3**
Drum-Buffer-Rope System

Line Balancing

- **Line balancing** is the assignment of work to stations in a line process so as to achieve the desired output rate with the smallest number of workstations.

- Normally, one worker is assigned to a station. Thus, the line that produces at the desired pace with the fewest workers is the most efficient one.

- It creates workstations with workloads as evenly balanced as possible. It seeks to create workstations so that the capacity utilization for the bottleneck is not much higher than for the other workstations in the line.

- Another difference is that line balancing applies only to line processes that do assembly work, or to work that can be bundled in many ways to create the jobs for each workstation in the line.

- Line balancing must be performed when a line is set up initially, when a line is rebalanced to change its hourly output rate, or when a product or process changes. The goal is to obtain work-stations with well-balanced workloads (e.g., every station takes roughly 3 minutes per customer in a cafeteria line with different food stations).

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Constructing a Precedence Diagram

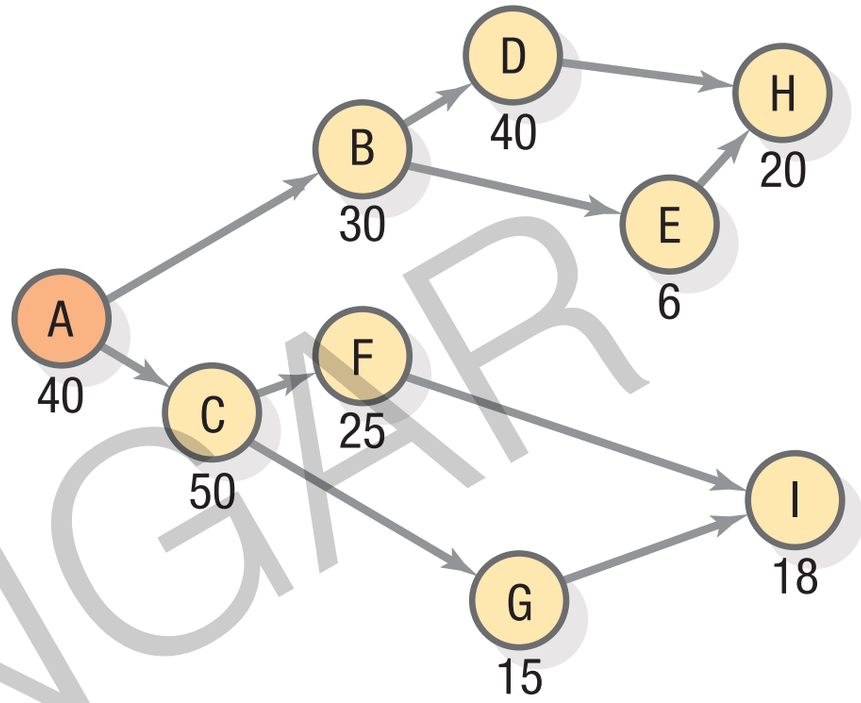
Green Grass, Inc., a manufacturer of lawn and garden equipment, is designing an assembly line to produce a new fertilizer spreader, the Big Broadcaster. Using the following information on the production process, construct a precedence diagram for the Big Broadcaster.

Work Element	Description	Time (sec)	Immediate Predecessor(s)
A	Bolt leg frame to hopper	40	None
B	Insert impeller shaft	30	A
C	Attach axle	50	A
D	Attach agitator	40	B
E	Attach drive wheel	6	B
F	Attach free wheel	25	C
G	Mount lower post	15	C
H	Attach controls	20	D, E
I	Mount nameplate	18	F, G
		Total 244	

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SOLUTION

Figure 7.4 shows the complete diagram. We begin with work element A, which has no immediate predecessors. Next, we add elements B and C, for which element A is the only immediate predecessor. After entering time standards and arrows showing precedence, we add elements D and E, and so on. The diagram simplifies interpretation. Work element F, for example, can be done anywhere on the line after element C is completed. However, element I must await completion of elements F and G.



DECISION POINT

Management now has enough information to develop a line-flow layout that clusters work elements to form workstations, with a goal being to balance the workloads and, in the process, minimize the number of workstations required.

- Precedence diagram is a diagram that allows one to visualize immediate predecessors better; work elements are denoted by circles, with the time required to perform the work shown below each circle.
- Work elements are the smallest units of work that can be performed independently.
- Immediate predecessors are work elements that must be done before the next element can begin.

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Calculating the Cycle Time, Theoretical Minimum and Efficiency

Green Grass's plant manager just received marketing's latest forecasts of Big Broadcaster sales for the next year. She wants its production line to be designed to make 2,400 spreaders per week for at least the next 3 months. The plant will operate 40 hours per week.

a. What should be the line's cycle time?

b. What is the smallest number of workstations that she could hope for in designing the line for this cycle time?

c. Suppose that she finds a solution that requires only five stations. What would be the line's efficiency?

SOLUTION

a. First, convert the desired output rate (2,400 units per week) to an hourly rate by dividing the weekly output rate by 40 hours per week to get $r = 60$ units per hour. Then, the cycle time is

$$c = 1/r = 1/60 \text{ (hour/unit)} = 1 \text{ minute/unit} = 60 \text{ seconds/unit}$$

b. Now, calculate the theoretical minimum for the number of stations by dividing the total time, Σt, by the cycle time, $c = 60$ seconds. Assuming perfect balance, we have

$$\text{TM} = \frac{\Sigma t}{c} = \frac{244 \text{ seconds}}{60 \text{ seconds}} = 4.067 \text{ or } 5 \text{ stations}$$

c. Now, calculate the efficiency of a five-station solution, assuming for now that one can be found:

$$\text{Efficiency (\%)} = \frac{\Sigma t}{nc}(100) = \frac{244}{5(60)}(100) = 81.3\%$$

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DECISION POINT

If the manager finds a solution with five stations that satisfies all precedence constraints, then that is the optimal solution; it has the minimum number of stations possible. However, the efficiency (sometimes called the *theoretical maximum efficiency*) will be only 81.3 percent. Perhaps the line should be operated less than 40 hours per week (thereby adjusting the cycle time) and the employees transferred to other kinds of work when the line does not operate.

- **Desired Output Rate:** The goal of line balancing is to match the output rate to the staffing or production plan. For example, if the plan calls for 4,800 units or customers per week and the line operates 80 hours per week, the desired output rate ideally would be 60 units or customers (4,800/80) per hour. Matching output to the plan ensures on-time delivery and prevents buildup of unwanted inventory or customer delays.

- **Cycle Time:** After determining the desired output rate for a line, the analyst can calculate the line's cycle time. A line's **cycle time** is the maximum time allowed for work on a unit at each station. If the time required for work elements at a station exceeds the line's cycle time, the station will be a bottleneck, preventing the line from reaching its desired output rate.

- **Theoretical Minimum (TM)** is a benchmark or goal for the smallest number of stations possible, where the total time required to assemble each unit (the sum of all work-element standard times) is divided by the cycle time.

- **Idle time** is the difference of (product of number of stations, n & cycle time, c) and total standard time, t required to assemble each unit.

- **Balance delay** is the amount by which efficiency falls short of 100 percent.

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Managerial Considerations

- In addition to balancing a line for a given cycle time, managers have four other considerations: (1) pacing, (2) behavioral factors, (3) number of models produced, and (4) different cycle times.

1. The movement of product from one station to the next as soon as the cycle time has elapsed is called **pacing**. Pacing manufacturing processes allows materials handling to be automated and requires less inventory storage area.

2. The most controversial aspect of line-flow layouts is **behavioral response**. Studies show that installing production lines increases absenteeism, turnover, and grievances. Paced production and high specialization (say, cycle times of less than 2 minutes) lower job satisfaction. One study even showed that productivity increased on unpaced lines.

3. A line that produces several items belonging to the same family is called a **mixed-model line**. In contrast, a single-model line produces one model with no variations. Mixed-model production enables a plant to achieve both high-volume production and product variety.

4. A line's **cycle time** depends on the desired output rate (or sometimes on the maximum number of workstations allowed). In turn, the maximum line efficiency varies considerably with the cycle time selected. Thus, exploring a range of cycle times makes sense.

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